

材料安全数据表

Material Safety Data Sheet

Sample name : Rechargeable Li-ion Battery

样 品 名 称 : 可充式锂离子电池组

C o n s i g n o r : Chongqing VDL Electronics Co.,Ltd.

委 托 单 位 : 重庆市紫建电子股份有限公司

东莞市全测电子科技有限公司
ATS Electronic Technology Co., Ltd.

东莞市全测电子科技有限公司
ATS Electronic Technology Co., Ltd.
广东省东莞市长安镇锦厦社区河东三路一号A栋三楼,
523852
3/F., Building A, No. 1, Hedong 3rd Road, Jinxia
Community, Changan, Dongguan, Guangdong, China

Tel: +86-769-38975958
Fax: +86-769-38975968
Web: www.dgats.com

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1. Identification of the product and supplier (产品和厂商信息)	
Name of goods 样品名称	Rechargeable Li-ion Battery 可充式锂离子电池组
Type/Mode 样品型号	682134 3.8V, 500mAh, 1.9Wh
Commissioned by 委托单位	Chongqing VDL Electronics Co., Ltd. 重庆市紫建电子股份有限公司
Commissioner address 委托单位地址	Building 1-4, Puli Industrial New Area, Zhaojia Street, Kaizhou District, Chongqing, China 重庆市开州区赵家街道浦里工业新区1-4号楼
Manufacturer 制造商	Chongqing VDL Electronics Co., Ltd. 重庆市紫建电子股份有限公司
Manufacturer address 制造商地址	Building 1-4, Puli Industrial New Area, Zhaojia Street, Kaizhou District, Chongqing, China 重庆市开州区赵家街道浦里工业新区1-4号楼
Inspection according to 鉴定依据	EEC Directive 93/112/EC 联合国《关于危险品货物运输的建议书》 UN "Recommendations on the TRANSPORT OF DANGEROUS GOODS"
Emergency telephone call 紧急联系电话	+86-755-29961201
接样日期/ Receiving date: 2019-12-18	
签发日期/ Issue date: 2020-01-01	

Tested by: 袁丁
主检: 袁丁

Reviewed by: 周丽
审核: 周丽

Approved by: 周丽
批准: 周丽



2. Composition/Information on Ingredient (成分/组成信息)		
Hazardous Ingredients (Chemical Name)	Concentration or concentration ranges (%)	CAS Number
Lithium Cobalt Oxide	30%~50%	12190-79-3
Conductive Additive	0.5%~1%	--
Polyvinylidene Fluoride(PVDF)	0.5%~1%	24937-79-9
Aluminium (Al)	5%~10%	7429-90-5
Graphite	15%~20%	7782-42-5
Styrene-Butadiene Rubber (SBR)	0.2%~1%	61789-96-6
Carboxymethylcellulose	0.2%~1%	9000-11-7
Copper (Cu)	5%~15%	7440-50-8
Nickel (Ni)	0.5%~1.5%	7440-02-0
Lithium Hexafluorophosphate	15%~25%	21324-40-3
Polyethylene	1%~5%	9002-88-4
Nylon	0.5%~1%	24937-16-4

3. Hazards Identification (主要危险性鉴定)	
爆炸危险性 Explosive risk	该物品不属于爆炸危险品 This article does not belong to the explosion dangerous goods
易燃危险性 Flammable risk	该物品不属于易燃危险品 This article does not belong to the flammable material
氧化危险性 Oxidation risk	该物品不属于氧化危险品 This article does not belong to the oxidation of dangerous goods
毒害危险性 Toxic risk	该物品不属于毒害危险品 This article does not belong to the toxic dangerous goods
放射危险性 Radioactive risk	该物品不属于放射危险品 This article does not belong to the radiation of dangerous goods
腐蚀危险性 Mordant risk	该物品不属于腐蚀危险品 This article does not belong to the corrosion of dangerous goods
其他危险性 other risk	该物品为可充式锂离子电池组，瓦时率为1.9Wh This article is Rechargeable Li-ion Battery, Watt hour rate 1.9Wh

4. First aid measures (急救措施)

Eye

Flush eyes with plenty of water for at least 15 minutes, occasionally lifting the upper and lower eyelids. Get medical aid.

Skin

Remove contaminated clothes and rinse skin with plenty of water or shower for 15 minutes. Get medical aid.

Inhalation

Remove from exposure and move to fresh air immediately. Use oxygen if available.

Ingestion

Give at least 2 glasses of milk or water. Induce vomiting unless patient is unconscious. Call a physician.

5. Fire-fighting measures (消防措施)

Flash Point: N/A.

Auto-Ignition Temperature: N/A.

Extinguishing Media: Water, CO₂.

Special Fire-Fighting Procedures

Self-contained breathing apparatus.

Unusual Fire and Explosion Hazards

Cell may vent when subjected to excessive heat-exposing battery contents.

Hazardous Combustion Products

Carbon monoxide, carbon dioxide, lithium oxide fumes.

6. Accidental release measures (泄漏应急处理)

Steps to be Taken in case Material is Released or Spilled

If the battery material is released, remove personnel from area until fumes dissipate. Provide maximum ventilation to clear out hazardous gases. Wipe it up with a cloth, and dispose of it in a plastic bag and put into a steel can. The preferred response is to leave the area and allow the battery to cool and vapors to dissipate. Provide maximum ventilation. Avoid skin and eye contact or inhalation of vapors. Remove spilled liquid with absorbent and incinerate.

Waste Disposal Method

It is recommended to discharge the battery to the end, to use up the metal lithium inside the battery, and to bury the discharged battery in soil.

7. Handling and storage (操作处置和储存)

The battery should not be opened, destroyed or incinerate, since they may leak or rupture and release to the environment the ingredients that they contain in the hermetically sealed container.

Do not short circuit terminals, or over charge the battery, forced over-discharge, throw to fire.

Do not crush or puncture the battery, or immerse in liquids.

Precautions to be taken in handling and storing

Avoid mechanical or electrical abuse. Storage preferably in cool, dry and ventilated area, which is subject to little temperature change. Storage at high temperatures should be avoided. Do not place the battery near heating equipment, nor expose to direct sunlight for long periods.

Other Precautions

The battery may explode or cause burns, if disassembled, crushed or exposed to fire or high temperatures. Do not short or install with incorrect polarity.

8. Exposure controls/personal protection (接触控制/个人防护)

Respiratory Protection

In case of battery venting, provide as much ventilation as possible. Avoid confined areas with venting cell cores. Respiratory Protection is not necessary under conditions of normal use.

Ventilation

Not necessary under conditions of normal use.

Other Protective Clothing or Equipment

Not necessary under conditions of normal use.

Personal Protection is recommended for venting battery

Respiratory Protection, Protective Gloves, Protective Clothing and safety glass with side shields.

9. Physical and chemical properties (物理和化学特性)

Appearance: Cuboid shape

Ref. No.: ATSU190305712

Odour: If leaking, smells of medical ether.

pH: Not applicable as supplied.

Flash Point: Not applicable unless individual components exposed.

Flammability: Not applicable unless individual components exposed.

Relative density: Not applicable unless individual components exposed.

Solubility (water): Not applicable unless individual components exposed.

Solubility (other): Not applicable unless individual components exposed.

10. Stability and reactivity (稳定性和反应活性)

Stability: Product is stable under conditions described in Section 7.

Conditions to avoid: Heat above 70°C or incinerate. Deform. Mutilate. Crush. Disassemble. Overcharge. Short circuit. Expose over a long period to humid conditions.

Materials to avoid: Oxidising agents, alkalis, water.

Hazardous Decomposition Products: Toxic Fumes, and may form peroxides.

Hazardous Polymerization: N/A.

If leaked, forbidden to contact with strong oxidizers, mineral acids, strong alkalies, halogenated hydrocarbons.

11. Toxicological information (毒理性资料)

Signs & symptoms: None, unless battery ruptures.

In the event of exposure to internal contents, vapour fumes may be very irritating to the eyes and skin.

Inhalation: Lung irritant.

Skin contact: Skin irritant

Eye contact: Eye irritant

Ingestion: Poisoning if swallowed

Medical conditions generally aggravated by exposure: In the event of exposure to internal contents, moderate to server irritation, burning and dryness of the skin may occur, Target organs nerves, liver and kidneys.

12. Ecological information (生态学资料)

Mammalian effects: None known at present.

Eco-toxicity: None known at present.

Bioaccumulation potential: Slowly Bio-degradable.

Environmental fate: None known environmental hazards at present.

13. Disposal consideration (废弃处置)

Do not incinerate, or subject cells to temperature in excess of 70°C, Such abuse can result in loss of seal leakage, and/or cell explosion. Dispose of in accordance with appropriate local regulations.

14. Transport information (运输信息)

Label for conveyance: Lithium Battery Label.

UN Number: UN3480 or UN3481

Packing Group: N/A

EmS No: 4.1-06

Marine pollutant: No

Proper Shipping name: 1) Lithium ion batteries; 2) Lithium ion batteries packed with equipment; 3) Lithium ion batteries contained in equipment. (including Lithium ion polymer batteries)

Hazard Classification: The goods shall be complied with the requirements of Section II (or Section IB) of Packing Instructions 965~967 of 61st DGR Manual of IATA (2020 edition), including the passing of the UN38.3 test. And also complies with the Special Provision 188 and 230 of IMDG CODE (Amdt 39-18) Edition.

15. Regulation information (法规信息)

Law information

《Dangerous Goods Regulations》

《Recommendations on the Transport of Dangerous Goods Model Regulations》

《International Maritime Dangerous Goods》

《Technical Instructions for the Safe Transport of Dangerous Goods》

《Classification and code of dangerous goods》

《Occupational Safety and Health Act》 (OSHA)

《Toxic Substance Control Act》 (TSCA)

《Consumer Product Safety Act》 (CPSA)

《Federal Environmental Pollution Control Act》 (FEPCA)

《The Oil Pollution Act》 (OPA)

《Superfund Amendments and Reauthorization Act TitleIII (302/311/312/313)》 (SARA)

《Resource Conservation and Recovery Act》 (RCRA)

《Safety Drinking Water Act》 (CWA)

《California Proposition 65》

《Code of Federal Regulations》 (CFR)

In accordance with all Federal, State and local laws.

16. Other information (其他信息)

This information is not effective to all the batteries manufactured by Chongqing VDL Electronics Co.,Ltd. This information comes from reliable sources, but no warranty is made to the completeness and accuracy of information contained. ATS Electronic Technology Co., Ltd. doesn't assume responsibility for any damage or loss because of misuse of batteries. User's should grasp the correct use method and be responsible for the use of batteries.